

VARUN K.V

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PROFESSIONAL SUMMARY

Computer Science graduate who has shipped production backend features at a funded EdTech startup, built and deployed an LLM-powered research tool with hallucination-controlled structured output, and trained deep learning models on real-world medical datasets. Proficient in Python, Ruby on Rails, FastAPI, GraphQL, Sidekiq, prompt engineering, and CNN-based computer vision. Brings production-grade code quality, multi-vendor integration experience, and a strong foundation in AI/ML — ready to build scalable backend and AI-powered applications.

WORK EXPERIENCE

Backend Developer Intern | Learnyst (EdTech Platform)

Bangalore, Karnataka | Feb 2026 - May 2026

- Integrated **2 third-party email validation vendors** behind a unified adapter using the **Strategy design pattern**, enabling providers to be swapped without modifying business logic - reducing future vendor onboarding effort to a single class addition.
- Migrated bulk email validation from synchronous to asynchronous processing using **Sidekiq and Redis**, eliminating admin dashboard blocking and enabling large recipient lists to be processed in the background while maintaining UI responsiveness.
- Built a **Sidekiq-driven reporting pipeline** generating deliverability reports (invalid rate, bounce metrics, deliverability score) auto-emailed to admins; exposed real-time validation status via **GraphQL APIs**, reducing dashboard over-fetching.
- Wrote **RSpec** tests with code coverage tracking, ensuring reliability of validation flows, webhook callbacks, and reporting logic across foreground and background job execution.

PROJECTS

Earnings Call Research Tool

Python - Streamlit - LLM API - Prompt Engineering - Docker - OCR | github.com/varun1-code/earnings-call-research-tool

- Built a structured **LLM-powered** research tool analyzing earnings call transcripts - extracting management tone, confidence level, forward guidance (revenue/margin/capex), and growth initiatives with **quote-grounded extraction** to eliminate hallucination.
- Multi-format ingestion pipeline (PDF, TXT, DOCX) with **Tesseract OCR** fallback; integrated **OpenAI-compatible LLM API** (NVIDIA-hosted); containerized and deployed via **Docker** with analyst-ready JSON/PDF export.

Trupti Yoga and Nature Cure - Full-Stack Membership Web Application

React 19 - FastAPI - SQLite - REST API - JWT - Pytest | [Live](#) | github.com/varun1-code/trupti-yoga-website

- Production-ready membership platform - **React 19 + FastAPI** backend with SQLite - handling user registration, UPI payment verification, admin approval workflow, and real-time Google Meet session access.
- Implemented **JWT authentication** with SHA-256 hashing, role-based admin panel, real-time IST-aware session status (auto-refresh every 30 s), and **53 unit tests** (Pytest) covering auth, payment, and timezone logic.

Skin Disease Detection - Deep Learning (Computer Vision)

PyTorch - CNN - Transfer Learning - HAM10000 - Derm7pt - Scikit-Learn | github.com/varun1-code/Skin-diseases-detection-

- Trained **CNN-based multi-class classification** models on HAM10000 + Derm7pt datasets (10K+ labeled images, 7 disease classes); applied data augmentation, normalization, and **weighted sampling** for class imbalance.
- Used **transfer learning** for improved generalization; built end-to-end training and real-time inference pipeline — achieved **85%+ validation accuracy** on 10K+ dermoscopic images across 7 disease classes.

TECHNICAL SKILLS

AI / GenAI: Prompt Engineering, HuggingFace Transformers, OpenAI-Compatible APIs, LLM Application Development, Structured Output Generation

Machine Learning: PyTorch, TensorFlow, Keras, Scikit-learn, Computer Vision, CNNs, Transfer Learning, Model Evaluation, Data Preprocessing

Backend & APIs: Python, FastAPI, Flask, Ruby on Rails, REST APIs, GraphQL, Redis, Sidekiq, Webhooks, JWT Authentication

Data & Frontend: SQL, PostgreSQL, SQLite, Pandas, NumPy, Power BI, React, Tailwind CSS, Streamlit

DevOps & Tools: Docker, Git, GitHub, Linux, Postman, Pytest, RSpec

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

EDUCATION

Maharaja Institute of Technology, Mysore | B.E. Computer Science Engineering | CGPA: 8.54 | Jun 2026

BGS PU College | Pre-University (PUC) | 94.36% | Jul 2021

CKS English School, Hassan | SSLC | 91% | Apr 2019

CERTIFICATIONS

Certifications: IBM AI Fundamentals (2025) | AI & Data Science - TejasPro (2025) | Python Full Stack - Pentagon Space (2026) | Backend Intern - Learnyst (2026)